

Note: This document only contains the graded problems viz Exercise 1 and 3. To view my solutions to the full Series 4 problem set, [click here](#).

Exercise 1

The blood groups of human beings are determined by the presence/absence of two different antigens, A and B. The following chart shows the determination of the four blood groups, A, B, AB and O, depending on the antigens.

	Antigen A present	Antigen A absent
Antigen B present	Blood group AB	Blood group B
Antigen B absent	Blood group A	Blood group O

Let p_A , p_B , p_{AB} and p_O denote the probabilities of a (randomly selected) person having blood group A, B, AB, or O, respectively. We select 500 people at random and note down their blood groups.

- (a) Formulate a statistical model for this data.
- (b) Let N_A , N_B , N_{AB} and N_O be the number of people (out of 500) having blood groups A, B, AB and O, respectively. Show that N_A , N_B , N_{AB} , N_O are jointly sufficient for p_A , p_B , p_{AB} , p_O .
- (c) Let q_A denote the probability of the antigen A being present. Similarly, q_B denotes the probability of the antigen B being present. Moreover, we believe that the presence/absence of the two antigens do not affect each other. How would you modify your model to take this additional piece of information into account? Also, find sufficient statistics for q_A and q_B in this updated model.
- (d) Are the sufficient statistics in parts (b) and (c) minimal sufficient? If not, then identify minimal sufficient statistics for the parameters in the two models.

Solution:

(a) Statistical Model Formulation:

Let $N = 500$ be the total number of randomly selected people. Each person falls into exactly one of four mutually exclusive blood group categories: A, B, AB, or O, with probabilities p_A, p_B, p_{AB} , and p_O respectively. Since the people are randomly selected, the trials are independent.

Thus, the random vector (N_A, N_B, N_{AB}, N_O) follows a *Multinomial distribution*.

We can formally define the statistical model as a triplet $(\mathcal{X}, \Omega, \mathcal{P})$:

- **Sample Space ($\mathcal{X}$):** The set of all possible frequency counts that sum up to $N = 500$.

$$\mathcal{X} = \{(n_A, n_B, n_{AB}, n_O) \in \{0, 1, \dots, 500\}^4 \mid n_A + n_B + n_{AB} + n_O = 500\}$$

- **Parameter Space (Ω):** The set of all valid probability vectors for the four categories.

$$\Omega = \{(p_A, p_B, p_{AB}, p_O) \in (0, 1)^4 \mid p_A + p_B + p_{AB} + p_O = 1\}$$

- **Family of Distributions ($\mathcal{P}$):** The set of Multinomial probability mass functions parameterized by $\mathbf{p} = (p_A, p_B, p_{AB}, p_O) \in \Omega$.

$$\mathcal{P} = \{\text{Multinomial}(500, \mathbf{p}) \mid \mathbf{p} \in \Omega\}$$

The probability mass function (pmf) for any observation $\mathbf{n} = (n_A, n_B, n_{AB}, n_O) \in \mathcal{X}$ is given by:

$$f(\mathbf{n}; \mathbf{p}) = P(N_A = n_A, N_B = n_B, N_{AB} = n_{AB}, N_O = n_O) = \frac{500!}{n_A! n_B! n_{AB}! n_O!} p_A^{n_A} p_B^{n_B} p_{AB}^{n_{AB}} p_O^{n_O}$$

(b) **Joint Sufficiency:**

Let the raw data be $X_1, X_2, \dots, X_N$ (where $N = 500$), representing the observed blood group of each randomly selected individual. Each X_i takes a value in the set $\{A, B, AB, O\}$.

Because the individuals are randomly selected, the observations are independent and identically distributed. The joint probability mass function of the entire sample is the product of the individual marginal probabilities:

$$f(\mathbf{x}; \mathbf{p}) = P(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N) = \prod_{i=1}^N P(X_i = x_i)$$

Instead of multiplying out 500 individual terms, we can group them by category. By definition, N_A, N_B, N_{AB} , and N_O are the frequency counts of each blood group in the sample. Thus, we can rewrite the joint pmf algebraically as:

$$f(\mathbf{x}; \mathbf{p}) = p_A^{N_A} p_B^{N_B} p_{AB}^{N_{AB}} p_O^{N_O}$$

Now, we apply the **Neyman-Fisher Factorization Theorem**. Let our joint statistic be $T(\mathbf{X}) = (N_A, N_B, N_{AB}, N_O)$ and our parameter vector be $\mathbf{p} = (p_A, p_B, p_{AB}, p_O)$. We can choose our functions as follows:

$$g(T(\mathbf{x}); \mathbf{p}) = p_A^{N_A} p_B^{N_B} p_{AB}^{N_{AB}} p_O^{N_O}$$

$$h(\mathbf{x}) = 1$$

Notice that the function g depends on the raw data $\mathbf{x}$ strictly through the statistic $T(\mathbf{x})$, and the function $h(\mathbf{x})$ is completely free of the parameter vector $\mathbf{p}$. Therefore, by the Factorization Theorem, the counts (N_A, N_B, N_{AB}, N_O) are jointly sufficient for the probabilities (p_A, p_B, p_{AB}, p_O) .

(c) **Updated Model and Sufficient Statistics:**

Given that the presence or absence of Antigen A and Antigen B are independent, we can express the probabilities of the four blood groups in terms of the marginal probabilities q_A and q_B :

- Blood group AB (both present): $p_{AB} = q_A q_B$
- Blood group A (A present, B absent): $p_A = q_A(1 - q_B)$
- Blood group B (A absent, B present): $p_B = (1 - q_A)q_B$
- Blood group O (both absent): $p_O = (1 - q_A)(1 - q_B)$

Modified Model: The model is still based on a Multinomial distribution of the counts, but the parameter space is now restricted to two parameters, $\mathbf{q} = (q_A, q_B) \in (0, 1)^2$. The probabilities p_A, p_B, p_{AB}, p_O are now functions of $\mathbf{q}$ as defined above.

Sufficient Statistics: To find the sufficient statistics for q_A and q_B , we return to the joint pmf of the raw sample $\mathbf{X} = (X_1, \dots, X_N)$ and substitute the new probabilities:

$$\begin{aligned} f(\mathbf{x}; q_A, q_B) &= p_A^{N_A} p_B^{N_B} p_{AB}^{N_{AB}} p_O^{N_O} \\ &= [q_A(1 - q_B)]^{N_A} [(1 - q_A)q_B]^{N_B} [q_A q_B]^{N_{AB}} [(1 - q_A)(1 - q_B)]^{N_O} \end{aligned}$$

Now, distribute the exponents and group the terms with base q_A and $(1 - q_A)$ together, and the terms with base q_B and $(1 - q_B)$ together:

$$\begin{aligned} f(\mathbf{x}; q_A, q_B) &= q_A^{N_A} (1 - q_B)^{N_A} (1 - q_A)^{N_B} q_B^{N_B} q_A^{N_{AB}} q_B^{N_{AB}} (1 - q_A)^{N_O} (1 - q_B)^{N_O} \\ &= q_A^{N_A + N_{AB}} (1 - q_A)^{N_B + N_O} q_B^{N_B + N_{AB}} (1 - q_B)^{N_A + N_O} \end{aligned}$$

Let $T_1 = N_A + N_{AB}$ (the total number of people with Antigen A) and $T_2 = N_B + N_{AB}$ (the total number of people with Antigen B). Notice that $N_B + N_O = N - T_1$ and $N_A + N_O = N - T_2$. We can rewrite the joint pmf as:

$$f(\mathbf{x}; q_A, q_B) = \underbrace{q_A^{T_1} (1 - q_A)^{N - T_1} q_B^{T_2} (1 - q_B)^{N - T_2}}_{g(T_1, T_2; q_A, q_B)} \times \underbrace{1}_{h(\mathbf{x})}$$

By the Neyman-Fisher Factorization Theorem, $T_1 = N_A + N_{AB}$ and $T_2 = N_B + N_{AB}$ are **jointly sufficient** statistics for (q_A, q_B) . Furthermore, since the joint pmf factors into a piece solely depending on T_1 and q_A , and another solely depending on T_2 and q_B , T_1 is sufficient for q_A , and T_2 is sufficient for q_B .

(d) **Minimal Sufficiency:**

To check for minimal sufficiency, we will apply the **Lehmann-Scheffé Theorem**. For two sample outcomes $\mathbf{x}$ and $\mathbf{y}$, we analyze the ratio of their likelihood functions: $\frac{f(\mathbf{x}; \theta)}{f(\mathbf{y}; \theta)}$. The statistic $T(\mathbf{X})$ is minimal sufficient if this ratio is constant (independent of the parameters) *if and only if* $T(\mathbf{x}) = T(\mathbf{y})$.

Analysis for Model (b): Here, our parameters are restricted by $p_O = 1 - p_A - p_B - p_{AB}$, and our sample counts are restricted by $n_O = N - n_A - n_B - n_{AB}$ where

$N = 500$. Let's rewrite the likelihood strictly in terms of the free parameters p_A, p_B , and p_{AB} :

$$\begin{aligned}\frac{f(\mathbf{x}; \mathbf{p})}{f(\mathbf{y}; \mathbf{p})} &= \frac{p_A^{n_A(\mathbf{x})} p_B^{n_B(\mathbf{x})} p_{AB}^{n_{AB}(\mathbf{x})} p_O^{N-n_A(\mathbf{x})-n_B(\mathbf{x})-n_{AB}(\mathbf{x})}}{p_A^{n_A(\mathbf{y})} p_B^{n_B(\mathbf{y})} p_{AB}^{n_{AB}(\mathbf{y})} p_O^{N-n_A(\mathbf{y})-n_B(\mathbf{y})-n_{AB}(\mathbf{y})}} \\ &= \left(\frac{p_A}{p_O}\right)^{n_A(\mathbf{x})-n_A(\mathbf{y})} \left(\frac{p_B}{p_O}\right)^{n_B(\mathbf{x})-n_B(\mathbf{y})} \left(\frac{p_{AB}}{p_O}\right)^{n_{AB}(\mathbf{x})-n_{AB}(\mathbf{y})}\end{aligned}$$

For this ratio to be independent of $\mathbf{p}$, the exponents must equal zero. This happens *if and only if*:

$$n_A(\mathbf{x}) = n_A(\mathbf{y}), \quad n_B(\mathbf{x}) = n_B(\mathbf{y}), \quad \text{and} \quad n_{AB}(\mathbf{x}) = n_{AB}(\mathbf{y})$$

This implies that the 3-dimensional statistic (N_A, N_B, N_{AB}) is minimal sufficient. The original 4-dimensional statistic (N_A, N_B, N_{AB}, N_O) is **not minimal** because N_O is redundant (since $N_O = 500 - N_A - N_B - N_{AB}$).

Minimal sufficient statistic for (b): (N_A, N_B, N_{AB}) (or any 3 of the 4 counts).

Analysis for Model (c): Let's analyze the likelihood ratio using the probability mass function derived in part (c), parameterized by (q_A, q_B) with statistics $t_1 = n_A + n_{AB}$ and $t_2 = n_B + n_{AB}$:

$$\begin{aligned}\frac{f(\mathbf{x}; q_A, q_B)}{f(\mathbf{y}; q_A, q_B)} &= \frac{q_A^{t_1(\mathbf{x})} (1 - q_A)^{N-t_1(\mathbf{x})} q_B^{t_2(\mathbf{x})} (1 - q_B)^{N-t_2(\mathbf{x})}}{q_A^{t_1(\mathbf{y})} (1 - q_A)^{N-t_1(\mathbf{y})} q_B^{t_2(\mathbf{y})} (1 - q_B)^{N-t_2(\mathbf{y})}} \\ &= \left(\frac{q_A}{1 - q_A}\right)^{t_1(\mathbf{x})-t_1(\mathbf{y})} \left(\frac{q_B}{1 - q_B}\right)^{t_2(\mathbf{x})-t_2(\mathbf{y})}\end{aligned}$$

For this ratio to be independent of q_A and q_B , the exponents must evaluate to zero. This happens *if and only if*:

$$t_1(\mathbf{x}) = t_1(\mathbf{y}) \quad \text{and} \quad t_2(\mathbf{x}) = t_2(\mathbf{y})$$

Therefore, the statistics found in part (c) **are minimal sufficient**.

Minimal sufficient statistic for (c): $(T_1, T_2) = (N_A + N_{AB}, N_B + N_{AB})$.

Exercise 3

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$.

- Find the joint density of $\frac{X_{(1)}}{X_{(n)}}$ and $X_{(n)}$.
Hint: Start with the joint pdf of $X_{(1)}$ and $X_{(n)}$, and use the Jacobian formula.
- Find the marginal distributions of $\frac{X_{(1)}}{X_{(n)}}$ and $X_{(n)}$. Verify that $\frac{X_{(1)}}{X_{(n)}}$ and $X_{(n)}$ are independent.

- (c) Verify that the distribution of $\frac{X_{(1)}}{X_{(n)}}$ is free of θ . Hence, deduce that $\frac{X_{(1)}}{X_{(n)}}$ is an ancillary statistic.
- (d) Repeat (a)–(c) with $X_{(i)}$ and $X_{(j)}$ for $1 \leq i < j \leq n$. Show that $\frac{X_{(i)}}{X_{(j)}}$ and $X_{(j)}$ are independent, and $\frac{X_{(i)}}{X_{(j)}}$ is ancillary.

Solution:

(a) Finding Joint Density:

We'll start by considering the joint pdf of $X_{(1)}$ and $X_{(n)}$. Thus, we have the following

$$f_{X_{(1)}, X_{(n)}}(x_1, x_n) = \frac{n(n-1)}{\theta^n} (x_n - x_1)^{n-2} \quad \forall 0 < x_1 < x_n < \theta$$

Now, to find the joint pdf of $X_{(1)}/X_{(n)}$ and $X_{(n)}$, we'll consider the following transformation in $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that

$$(X_{(1)}, X_{(n)}) \mapsto \left(\frac{X_{(1)}}{X_{(n)}}, X_{(n)} \right)$$

Let $X_{(1)}/X_{(n)} = U$ and $X_{(n)} = V$. This implies $x_1 = uv$ and $x_n = v$. Since we can uniquely identify x_1 and x_n given u, v , we have a bijective map and hence can use the Jacobian formula. We'll start by computing the determinant of the Jacobian

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_n}{\partial u} & \frac{\partial x_n}{\partial v} \end{pmatrix} = \begin{pmatrix} v & u \\ 0 & 1 \end{pmatrix}$$

Thus, we have $\det(J) = v$. Now we'll define the support for U, V . Thus, we have the following

$$0 < x_1 < x_n < \theta \implies 0 < uv < v < \theta$$

This implies,

$$0 < v < \theta$$

$$0 < u < 1$$

We have the following formula joint pdf

$$f_{U,V}(u, v) = f_{X_{(n)}, X_{(1)}}(x_1(u, v), x_n(u, v)) \cdot \det(J)$$

Thus, we have the following joint pdf for U, V

$$\begin{aligned} f_{U,V}(u, v) &= \left[\frac{n(n-1)}{\theta^n} (v - uv)^{n-2} \right] v \\ &= \left[\frac{n}{\theta^n} v^{n-1} \right] [(n-1)(1-u)^{n-2}] \end{aligned}$$

$$\boxed{f_{U,V}(u, v) = \left[\frac{n}{\theta^n} v^{n-1} \right] [(n-1)(1-u)^{n-2}] \quad \forall 0 < v < \theta \text{ and } 0 < u < 1}$$

(b) **Finding Marginal Density:**

We'll use the joint pdf of U, V derived in the previous step to find the marginals

- For $U = X_{(1)}/X_{(n)}$. Consider the following

$$\begin{aligned} f_U(u) &= \int_0^\theta f_{U,V}(u, v) dv \\ &= (n-1)(1-u)^{n-2} \int_0^\theta \frac{n}{\theta^n} v^{n-1} dv \\ &= (n-1)(1-u)^{n-2} \underbrace{\left[\frac{n}{\theta^n} \frac{v^n}{n} \right]_0^\theta}_1 \end{aligned}$$

Thus we have

$$\boxed{f_U(u) = (n-1)(1-u)^{n-2} \quad \forall 0 < u < 1}$$

- For $V = X_{(n)}$. Consider the following

$$\begin{aligned} f_V(v) &= \int_0^1 f_{U,V}(u, v) du \\ &= \frac{n}{\theta^n} v^{n-1} \underbrace{\int_0^1 (n-1)(1-u)^{n-2} du}_{=1 \text{ as this is pdf of } U} \end{aligned}$$

Thus we have

$$\boxed{f_V(v) = \frac{n}{\theta^n} v^{n-1} \quad \forall 0 < v < \theta}$$

We can clearly see that the joint pdf of U, V is just the product of the marginals. This clearly indicates that U and V are independent

(c) **Proving Ancillarity for U :**

We obtained the following density for U

$$f_U(u) = (n-1)(1-u)^{n-2} \quad \forall 0 < u < 1$$

We define any *ancillary statistic* as a statistic whose distribution is independent of the parameter. As it happens for U , we can clearly see that the density of U does not have the term of θ thus making U an ancillary statistic for θ . Thus we have

$$U = \frac{X_{(1)}}{X_{(n)}}$$

is an ancillary statistic for θ

(d) **Repeting for any $X_{(i)}, X_{(j)}$**

Now, we'll be repeting the same exercises for $X_{(i)}$ and $X_{(j)}$. We'll again start by writing the joint pdf of $X_{(i)}, X_{(j)}$. We have the following joint pdf for any distribution family

$$f_{X_{(i)}, X_{(j)}}(x_i, x_j) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} [F(x_i)]^{i-1} [F(x_j) - F(x_i)]^{j-i-1} [1 - F(x_j)]^{n-j} f(x_i) f(x_j)$$

for $-\infty < x_i < x_j < \infty$.

Thus, for $\text{Uniform}(0, \theta)$, we have the following

$$f_{X_{(i)}, X_{(j)}}(x_i, x_j) = \begin{cases} \frac{n!}{(i-1)!(j-i-1)!(n-j)! \theta^n} x_i^{i-1} (x_j - x_i)^{j-i-1} (\theta - x_j)^{n-j} & \text{for } 0 < x_i < x_j < \theta \\ 0 & \text{otherwise} \end{cases}$$

As done in previous parts, we'll consider the following transformation from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. It is a bijective map as we can uniquely identify x_i, x_j given u, v

$$(X_{(i)}, X_{(j)}) \mapsto \left(\frac{X_{(i)}}{X_{(j)}}, X_{(j)} \right)$$

Let $X_{(i)}/X_{(j)} = U$ and $X_{(j)} = V$. This implies $x_i = uv$ and $x_j = v$. We'll start by computing the determinant of the Jacobian

$$J = \begin{pmatrix} \frac{\partial x_i}{\partial u} & \frac{\partial x_i}{\partial v} \\ \frac{\partial x_j}{\partial u} & \frac{\partial x_j}{\partial v} \end{pmatrix} = \begin{pmatrix} v & u \\ 0 & 1 \end{pmatrix}$$

Thus, we have $\det(J) = v$. Now we'll define the support for U, V . Thus, we have the following

$$0 < x_i < x_j < \theta \implies 0 < uv < v < \theta$$

This implies,

$$0 < v < \theta$$

$$0 < u < 1$$

We have the following formula joint pdf

$$f_{U,V}(u, v) = f_{X_{(i)}, X_{(j)}}(x_i(u, v), x_j(u, v)) \cdot \det(J)$$

Thus, we have the following joint pdf for U, V

$$\begin{aligned} f_{U,V}(u, v) &= \left[\frac{n!}{(i-1)!(j-i-1)!(n-j)! \theta^n} (uv)^{i-1} (v - uv)^{j-i-1} (\theta - v)^{n-j} \right] v \\ &= \frac{n!}{(i-1)!(j-i-1)!(n-j)! \theta^n} u^{i-1} v^{i-1} (1-u)^{j-i-1} v^{j-i-1} (\theta - v)^{n-j} v \\ &= \frac{n!}{(i-1)!(j-i-1)!(n-j)! \theta^n} u^{i-1} (1-u)^{j-i-1} v^{(i-1)+(j-i-1)+1} (\theta - v)^{n-j} \\ &= \left[\frac{(j-1)!}{(i-1)!(j-i-1)!} u^{i-1} (1-u)^{j-i-1} \right] \left[\frac{n!}{(j-1)!(n-j)! \theta^n} v^{j-1} (\theta - v)^{n-j} \right] \end{aligned}$$

Notice that the joint density factors perfectly into a product of two functions: one that depends solely on u and another that depends solely on v . Let us denote these factors as:

$$g(u) = \frac{(j-1)!}{(i-1)!(j-i-1)!} u^{i-1} (1-u)^{j-i-1}, \quad 0 < u < 1$$

$$h(v) = \frac{n!}{(j-1)!(n-j)! \theta^n} v^{j-1} (\theta-v)^{n-j}, \quad 0 < v < \theta$$

We can recognize $h(v)$ as the exact marginal probability density function of the j -th order statistic, $X_{(j)}$. Because $f_{U,V}(u, v) = g(u)h(v)$ and the joint support of (U, V) is a rectangle $((0, 1) \times (0, \theta))$, we can conclude two things simultaneously:

- U and V are independent.
- The marginal density of U is exactly $f_U(u) = g(u)$.

Thus, the marginal distribution of $U = \frac{X_{(i)}}{X_{(j)}}$ is a $\text{Beta}(i, j-i)$ distribution:

$$f_U(u) = \frac{(j-1)!}{(i-1)!(j-i-1)!} u^{i-1} (1-u)^{j-i-1} \quad \forall 0 < u < 1$$

Also, we have the marginal density for $V = X_{(j)}$ as the following:

$$h(v) = \frac{n!}{(j-1)!(n-j)! \theta^n} v^{j-1} (\theta-v)^{n-j}, \quad 0 < v < \theta$$

Finally, observe the marginal density $f_U(u)$. It depends only on the known constants i, j , and n . The unknown parameter θ has completely vanished from the density function. Because the distribution of U is free of θ , we deduce that $U = \frac{X_{(i)}}{X_{(j)}}$ is an **ancillary statistic**.